Health Insurance Coverage and Hypertension Awareness Among U.S. Adults

Survey-weighted analysis of NHANES 2017–2018

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Abstract

Background. Health insurance may facilitate screening and recognition of hypertension, but current insurance coverage and a self-reported previous diagnosis need not refer to the same point in time. We evaluated the association between insurance status and hypertension awareness among adults younger than Medicare eligibility and examined how analytical choices affect interpretation.

Methods. We analyzed the 2017–2018 National Health and Nutrition Examination Survey (NHANES), a multistage probability sample of the U.S. civilian noninstitutionalized population. Participants aged 20–64 years with operational hypertension, defined by mean systolic blood pressure (BP) at least 130 mmHg, mean diastolic BP at least 80 mmHg, or current prescribed antihypertensive medication, formed the study domain. The outcome was a self-reported previous professional diagnosis of hypertension. Current health insurance was the exposure. We estimated design-adjusted awareness prevalence and fitted survey-weighted logistic regression models incorporating age, sex, race/ethnicity, education, and income-to-poverty ratio. Prespecified and post-review sensitivity analyses evaluated definitions, missingness, and inferential assumptions.

Results. Of 1,759 adults with classifiable hypertension and observed insurance and awareness, 1,467 were insured and 292 uninsured. Weighted awareness was 55.1% (95% confidence interval [CI], 49.2–60.9) versus 43.1% (34.2–52.4), respectively. In the fully adjusted complete-case model (n = 1,534), the odds ratio comparing uninsured with insured adults was 0.60 (95% CI, 0.22–1.62). The model-standardized awareness difference was −11.6 percentage points (95% CI, −33.7 to 10.5). Estimates were imprecise and varied across hypertension definitions.

Conclusion. The observed difference was directionally consistent with previous hypertension-care research but did not establish a clear adjusted association. The analysis demonstrates the importance of explicit outcome definitions, design-based uncertainty, and sensitivity analysis in secondary epidemiological research.

Keywords: hypertension awareness; health insurance; NHANES; epidemiology; complex survey analysis; reproducibility

1. Introduction

Hypertension is a major contributor to cardiovascular morbidity. Recognition of elevated blood pressure is a prerequisite for many steps in clinical management, including follow-up, medication initiation, and monitoring. Yet measuring *awareness* is not the same as observing a documented diagnosis: a survey respondent may not recall being informed, may have received a different diagnostic label, or may have a newly elevated measurement at the examination. Population estimates must therefore state exactly what they measure.

Insurance can lower financial barriers to clinic visits and diagnostic services, but it is also correlated with employment, age, socioeconomic conditions, disability, and healthcare-seeking behavior. Furthermore, the insurance measure available in a cross-sectional survey describes current coverage, whereas awareness describes whether a professional *ever* reported hypertension. A previous diagnosis could precede or even affect current insurance eligibility. An association between the two measures cannot, by itself, identify the causal effect of gaining coverage.

Published NHANES analyses establish the broader research context. Egan and colleagues documented widening insurance-related differences in awareness, treatment, and control across earlier surveys [1]. In 2005–2010, awareness was 80.7% among insured adults with hypertension and 65.2% among uninsured adults, based on the study's population and hypertension definition [1]. Calixte and colleagues analyzed NHANES 2011–2018 and observed greater hypertension unawareness among adults lacking a routine place of healthcare [2]. These studies concern related but not identical exposures and target populations. Muntner and colleagues reported national trends in hypertension management using a 140/90-or-medication definition, illustrating how eligibility criteria and historical practice affect the denominator of awareness estimates [3].

We asked: **Among U.S. adults aged 20–64 with operationally defined hypertension, is lack of current insurance associated with lower self-reported hypertension awareness after demographic and socioeconomic adjustment?** The purpose was not to claim a previously unknown relationship. Instead, the analysis examined how a plausible clinical question can be addressed transparently using one public survey cycle, with explicit attention to survey design, measurement, model uncertainty, and robustness.

2. Methods

2.1 Study design and population

We conducted a retrospective, cross-sectional secondary analysis of public-use NHANES 2017–2018 data. The analysis combined demographics (DEMO_J), examination blood pressure (BPX_J), blood-pressure questionnaire items (BPQ_J), insurance (HIQ_J), and healthcare-utilization items (HUQ_J). Data were linked by the respondent sequence identifier (SEQN); one-to-one merges, identifier uniqueness, and provenance of raw files were checked in the source repository [4]. The separate 2017–March 2020 prepandemic release was not used.

Eligibility was restricted to examined adults aged 20–64 years with positive examination sampling weights and classifiable hypertension status. The upper age bound limits the abrupt insurance-coverage differences associated with Medicare eligibility at age 65; it does not eliminate differences among younger adults. Domain analyses preserved the original survey design before subpopulation restriction. Records with unknown exposure or awareness were excluded from analyses requiring those fields. Additional covariate exclusions are described below.

2.2 Blood pressure and hypertension classification

Mean systolic and diastolic BP were computed separately from valid available first, second, and third examination readings. A valid zero diastolic value was not automatically treated as missing. Operational hypertension was defined as mean systolic BP ≥130 mmHg, mean diastolic BP ≥80 mmHg, **or current use**

of prescribed antihypertensive medication. Medication status relied on the current-use item BPQ050A, not BPQ040A, which addresses whether a professional ever advised medication. Skip patterns were interpreted with reference to the original NHANES codebook, with unresolved responses retained as unknown [4].

A known elevated pressure component or positive current-medication response was sufficient for a positive classification. Negative classification required below-threshold measurements and no indicated medication use; otherwise ascertainment could remain unresolved. This logic allowed treated cases to be retained even without complete recorded BP. The classification is an epidemiological operational measure rather than proof of persistent, clinically diagnosed hypertension from repeated visits.

2.3 Exposure, outcome, and adjustment variables

The exposure contrasted currently uninsured with insured respondents using HIQ011. Awareness was a yes/no response to BPQ020, asking whether a physician or another health professional had ever told the participant they had hypertension. Refused, unknown, and inapplicable values were not recoded as negative answers. The main covariates were age per decade, sex, six race/ethnicity categories, four educational-attainment categories, and continuous family income-to-poverty ratio (PIR). These factors were selected because they may relate to healthcare access and diagnosis reporting; adjustment was not based on automated significance selection.

We did not adjust for usual place of healthcare in the primary insurance model, because healthcare access may lie on pathways connecting insurance to diagnosis. Usual care was examined separately as a secondary exposure. Income missingness and the potential temporal ambiguity of covariates were considered in sensitivity analyses. Race/ethnicity categories were interpreted as social measures rather than biological causal categories.

2.4 Complex survey estimation and regression

NHANES examination weights (WTMEC2YR), masked strata (SDMVSTRA), and masked primary sampling units (SDMVPSU) defined the sampling design. Taylor-series linearization was used for variances. The primary regression used the R **survey** package, with the complete survey design specified before defining the analytic domain [5]. Python handled data preparation and performed an independent weighted-score/Taylor calculation; reported confidence intervals were not derived from naïve weighted regression standard errors.

Three logistic models were fitted: Model 1 included only insurance; Model 2 added age and sex; Model 3 additionally included race/ethnicity, education, and PIR. Odds ratios (ORs) were exponentiated logistic coefficients, not relative risks. Each model used complete cases for its own predictors; common-sample analyses helped distinguish sample changes from adjustment. Coefficient intervals used the model's residual survey degrees of freedom. The fully adjusted model had only three residual reference degrees of freedom, making the uncertainty notably conservative and wide. Standardized predicted awareness probabilities were averaged over the weighted complete-case covariate distribution; their contrast was expressed as a percentage-point difference.

2.5 Robustness, missing data, and validation

Sensitivity analyses compared 130/80 with 140/90 thresholds, medication-inclusive with measured-elevated-only case definitions, alternative BP-reading rules, age ranges, and covariate specifications. A post-review assessment allowed current-medication cases without complete BP and retained the original BP-complete rule as a sensitivity. Results arising from post-review changes are exploratory rather than prospectively confirmed.

Income-to-poverty ratio was missing in a meaningful subset. The principal analysis used complete cases; a secondary response-weighting analysis modeled PIR observation using available demographic and clinical predictors and refitted nuisance and outcome models in stratified delete-one-PSU replicates. Its interpretation depends on missing-at-random and modeling assumptions and does not prove that missingness bias has been removed. Python/R coding and covariance outputs were compared using recorded tolerances; convergence and test checks were documented in the repository. This validation addresses computational consistency, not unmeasured confounding or the correctness of causal assumptions.

3. Results

3.1 Study sample and descriptive findings

NHANES 2017–2018 contained 9,254 respondents. Of 4,069 adults aged 20–64, 3,873 completed the relevant examination; 3,709 had classifiable hypertension status, and 1,767 met the operational hypertension definition. Eight had missing insurance or awareness data, leaving 1,759 participants for the main descriptive contrast. Model 3 used 1,534 covariate-complete observations, including 847 reporting a previous diagnosis and 687 not reporting one. Of the 1,759 descriptive participants, 1,467 were insured and 292 uninsured.

Survey-weighted awareness was 55.1% (95% CI, 49.2–60.9) among the insured and 43.1% (34.2–52.4) among the uninsured (Figure 1), an unadjusted difference of approximately 12 percentage points. Uninsured adults were younger on average (45.6 versus 48.7 years), had lower mean PIR (1.9 versus 3.3), and had higher measured mean systolic BP (137.1 versus 132.2 mmHg). Values for these characteristics use variable-specific denominators, and the descriptive comparison should not be mistaken for covariate-adjusted inference.

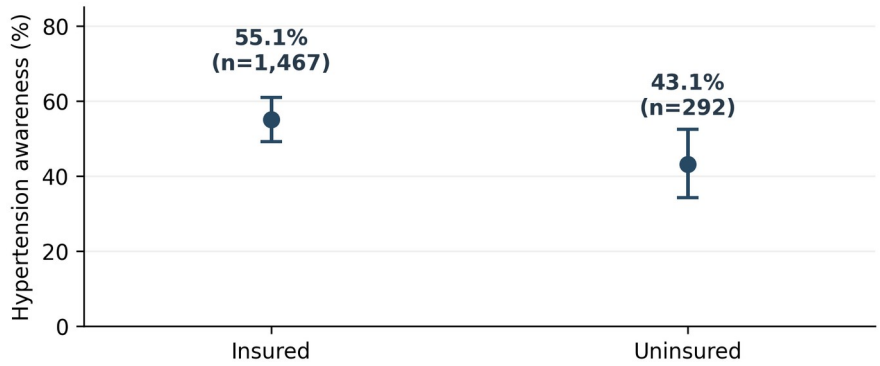


Figure 1. Survey-weighted hypertension awareness among insured and uninsured adults. Error bars show 95% design-based confidence intervals; sample sizes are unweighted.

3.2 Adjusted association and marginal probabilities

Model 1 estimated lower odds of reporting awareness among uninsured respondents (OR 0.62; 95% CI, 0.41–0.94). After age and sex adjustment, the OR was 0.69 (0.43–1.11). The fully adjusted Model 3 estimate was 0.60 (0.22–1.62), based on 1,534 participants. The interval included 1 and was compatible with substantially lower, similar, or potentially higher odds. It does not establish a statistically clear adjusted difference. Different models also used different complete-case samples; common-sample checks did not resolve this imprecision.

Model and covariates	Unweighted n	OR, uninsured vs insured (95% CI)	Residual survey df
Model 1: insurance	1,759	0.62 (0.41–0.94)	14
Model 2: + age and sex	1,759	0.69 (0.43–1.11)	12
Model 3: + race/ethnicity, education, PIR	1,534	0.60 (0.22–1.62)	3

Table 1. Survey-adjusted logistic models. OR = odds ratio; CI = confidence interval; PIR = income-to-poverty ratio. Degrees of freedom are the reference values used for the coefficient intervals.

Within the Model 3 weighted covariate distribution, standardized awareness probabilities were 54.7% (95% CI, 46.1–62.9) under insured status and 43.1% (25.2–63.0) under uninsured status. The adjusted probability difference was −11.6 percentage points (−33.7 to 10.5). These predictions describe a modeled contrast, not the effect of assigning insurance.

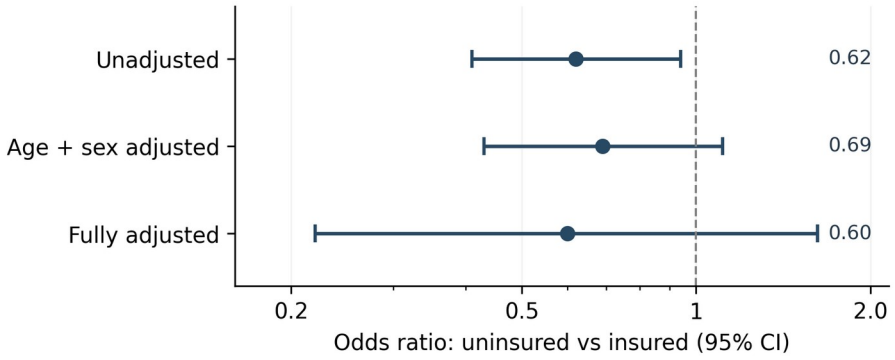


Figure 2. Odds ratios for the uninsured-versus-insured contrast across increasingly adjusted models (95% survey-based confidence intervals). A value of 1 denotes no association.

3.3 Sensitivity analyses and model diagnostics

Alternative definitions changed the size and composition of the hypertension domain. With the 140/90-or-medication definition, the adjusted OR was 0.78 (95% CI, 0.19–3.20; n = 998). With measured high BP only at 130/80, it was 0.67 (0.23–1.98; n = 1,279). At measured 140/90 alone, the point estimate reversed to 1.14 (0.25–5.23; n = 573). None of these intervals supported a firm conclusion. The difference across definitions is substantive because diagnosed and treated adults may enter the medication-inclusive group even if their observed BP is controlled.

Removing PIR from adjustment yielded an OR of 0.68 (0.37–1.25; n = 1,758). The response-weighted missing-PIR sensitivity yielded 0.66 (0.29–1.51; n = 1,534 with observed PIR contributing to the weighted fit), remaining inconclusive. A sensitivity adopting the full-design degrees of freedom while retaining the

primary coefficients gave 0.60 (0.31–1.17), also inconclusive. The original complete-BP restriction gave 0.59 (0.21–1.63). These comparisons were used to assess sensitivity, not to select the most favorable p-value.

Model 3 converged, but cells defined jointly by insurance and demographic categories were sometimes small; the smallest such cell contained 12 observations and as few as two aware responses. Sparse support, weight variability, and limited survey degrees of freedom warrant caution in interpreting highly adjusted estimates. Recorded numerical comparisons showed close agreement between independently implemented Python and R computations, including regression coefficients and Taylor covariance results. Numerical replication alone cannot ensure epidemiological validity.

Table 2. Selected robustness checks for the uninsured-versus-insured adjusted association. All intervals are 95% CIs; domains differ across case definitions.

Analysis specification	n	Adjusted OR (95% CI)
Primary, ≥130/80 or medication	1,534	0.60 (0.22–1.62)
≥140/90 or medication	998	0.78 (0.19–3.20)
Measured ≥130/80 only	1,279	0.67 (0.23–1.98)
Measured ≥140/90 only	573	1.14 (0.25–5.23)
No PIR covariate	1,758	0.68 (0.37–1.25)
PIR response-weighted sensitivity	1,534*	0.66 (0.29–1.51)
Complete BP restriction	1,503	0.59 (0.21–1.63)

**In the response-weighted sensitivity, 1,534 respondents with observed PIR contribute to the weighted fit; the modeled response target includes 1,758 outcome/exposure- and other-covariate-complete adults.*

4. Discussion

This analysis observed lower crude hypertension awareness among uninsured than insured adults aged 20–64 in NHANES 2017–2018. The point estimate remained below one after covariate adjustment, but the confidence interval was wide and included no association. Thus, the principal inference is uncertainty rather than confirmation of an insurance effect. Although the crude difference was approximately 12 percentage points, the estimated fully adjusted difference ranged from substantial disadvantage to a modest advantage for uninsured adults.

The direction of the crude contrast is compatible with earlier evidence. Egan et al. reported lower awareness among uninsured adults in prior NHANES cycles [1]. Calixte et al. showed that lacking a routine place of care was associated with greater unawareness using pooled 2011–2018 surveys [2]. However, routine healthcare access is not the same construct as current insurance, and the two studies' regression outcomes and population definitions are not directly interchangeable. Muntner et al. examined awareness and control under the older 140/90-or-medication case definition [3]. Their reported levels should not be compared numerically with the present lower-threshold sample as if the denominator were constant. These studies already establish the clinical relevance of healthcare access; this single-cycle exercise does not claim to discover that association anew.

Two measurement problems are especially important. First, the observed 130/80 threshold identifies some respondents who may never previously have met the diagnostic threshold used in clinical practice. Someone with one elevated examination reading and no recollection of a hypertension diagnosis may be classified as unaware even without demonstrating an omission in care. Second, medication-based case identification depends on questionnaire paths related to diagnosis. Treatment can normalize measured BP

yet increase a respondent's probability of inclusion through the medication branch. Conditioning on this operational case definition may therefore induce selection and definitional relationships among insurance, prior diagnosis, and treatment. Restricting cases to measured hypertension reduces one linkage but excludes well-controlled treated participants, producing a different population rather than a universally corrected estimate.

The statistical uncertainty also matters. A full complex-survey design can have more observations than independent sampling units. The conservative residual-degree-of-freedom convention left only three reference degrees of freedom in the largest adjusted model, inflating interval width. Using full-design degrees of freedom narrowed intervals but did not establish a clear adjusted association. Likewise, complete-case exclusions for PIR reduced the analytic sample; response weighting addressed an alternative missingness assumption but did not eliminate uncertainty or guarantee unbiasedness. These features argue against overinterpreting either the sign or apparent magnitude of the adjusted effect.

The design has important limitations. Current insurance was measured at the survey date, whereas previous diagnosis could have occurred years earlier, leaving temporal ordering and reverse causation unresolved. Awareness and medication use are self-reported; BP was measured at one examination, not across diagnostic visits. Unmeasured health literacy, prior coverage, geography, healthcare quality, and morbidity could confound associations. Weighting accounts for the published sampling design but cannot necessarily correct all examination nonresponse or item missingness. CDC has documented sampling-location and nonresponse concerns for the 2017–2018 cycle and recommends pooling compatible cycles where scientifically appropriate [6]. These estimates pertain to a historical population, not today's insurance environment.

The study's strengths lie in the reproducible analytical process: prespecified variables, retained unknown responses, explicit domain construction, defensible weighted inference, sensitivity to thresholds and case ascertainment, and transparent reporting of numerical checks. Its limitations are those of a single historical cross-section, a highly adjusted model with limited design degrees of freedom, and restricted novelty. A logical next step is a protocol-led comparison with Calixte et al.'s pooled-cycle healthcare-access analysis, followed by a justified extension assessing how different hypertension definitions influence estimated awareness disparities.

5. Conclusion

Among adults aged 20–64 meeting an operational NHANES definition of hypertension, uninsured respondents had lower crude reported awareness. The adjusted evidence was imprecise and sensitive to some choices about case selection, so the analysis cannot establish either a clear adjusted association or a causal effect of insurance. Its value as a research-training sample lies in using transparent population definitions, survey-aware statistics, independent computational checks, and appropriately restrained interpretation.

Notes on data, ethics, and contribution

This is a condensed academic writing sample derived from a public-data research-training repository. The source project documents AI-assisted coding and writing; any applicant using the sample should verify computations and describe their actual contributions accurately. No publication, independent supervision, or IRB exemption is asserted. NHANES obtained study-level informed consent and ethics review through its own procedures [4]. Tables and figures reproduce analyses recorded in the supplied project; this document does not represent a newly executed study.

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